

Functional traits shape tree growth response to winter freeze-thaw cycle and neighborhood crowding in humid temperate forests

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ABSTRACT

Climate change is intensifying the occurrence of hydric events, such as summer water availability and winter freeze-thaw cycles, which are increasingly significant in temperate regions. These events pose substantial threats to tree hydraulic functions and thereby limiting tree growth. In this study, we investigated the compound effects of climate-induced stressors and neighborhood crowding on tree growth, and tested how species functional traits mediate these responses. We combined annual growth data over 8 years from 593 individuals representing 20 tree species in Northeast China to evaluate the relative influence of water availability, freeze-thaw cycles, and their interaction with local crowding on tree growth, and how species functional traits mediate these responses. Our findings indicate that tree growth declined with increasing freeze-thaw cycles frequency, whereas summer water availability had no detectable effect. Tree growth was limited by neighborhood crowding, which appeared to operate largely independently of freeze-thaw cycles. In addition, species with higher xylem hydraulic efficiency, lower wood density and lower specific leaf area grew faster and were more sensitive to freeze-thaw cycles, while species with higher xylem hydraulic efficiency were less sensitive to neighborhood crowding. Our results demonstrate distinct and independent roles of freeze-thaw cycles and neighborhood crowding in shaping temperate tree growth, suggesting that considering the freeze-thaw cycles may improve predictions of temperate forest dynamics facing altered climate changes. Furthermore, species traits can capture how temperate trees cope with different stressors, highlighting the importance of integrating functional traits for a more comprehensive understanding of tree responses to environmental stressors.

1. Introduction

Tree growth is a key demographic process that drives forest dynamics (Rüger et al., 2020) and impacts vital ecosystem function, such as carbon sequestration, nutrient cycling, and climate regulation (Migliavacca et al., 2021). Recent trends show an increased frequency and intensity of extreme climate events, which are significantly impacting tree growth and contributing to uncertainties about the future of forests globally (Anderegg et al., 2020; Bauman et al., 2022). Notably, hydraulic dysfunction resulting from water stress is a crucial physiological process that leads to a decline in tree growth (Liu et al., 2018; Arend et al., 2021; Shovon et al., 2024). In temperate forests, this water

stress manifests not only through low water availability in summers but also via freeze-thaw cycles (FTC) in winters (Améglio et al., 2002; Oliveira et al., 2021). The interplay and relative contributions of these distinct seasonal water-related stressors to tree growth remain poorly understood, emphasizing the need for studies that dissect these dynamics under changing climatic conditions.

At the regional scale, seasonal variations in temperature and precipitation in temperate ecosystems determine soil water availability in summers and winters, each leading to distinct forms of hydraulic dysfunction (Mayr and Sperry, 2010; Choat et al., 2012; Niu et al., 2017). During periods of low soil water availability, persistent water deficit results in decreased stem water potential, which can lead to

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xylem embolism, a condition where air blocks the water transport system within the tree (Borghetti et al., 1998). During cold winter months, the lowering temperatures can cause the sap within the tree's xylem to freeze, reducing the solubility of gases and resulting the formation of cavitation. These cavitation events might contract during freezing, but as they thaw with rising spring temperatures, larger cavitation can expand, potentially leading to the development of xylem embolisms (Sperry and Tyree, 1988; Tyree and Sperry, 1989). This process impairs the stem hydraulic system, increasing resistance to water transport from roots to leaves and generally diminishing growth throughout the growing season (Zhang et al., 2015). While many studies in temperate forests have focused on the role of drought-induced hydraulic dysfunction influences tree growth (Hillbrand et al., 2016; Santiago et al., 2018), fewer research has explored the impact of FTC on tree growth (Choat et al., 2011; Niu et al., 2017). To date, no study so far has comprehensively assessed how interplay between summer drought and winter FTC effects tree growth in temperate forests.

At the local scale, neighborhood interactions can significant modifiers of tree growth (Canham et al., 2006; Kunstler et al., 2011). Neighboring trees often compete for shared limiting resources such as light, water and nutrients, and may even face common biotic pressures such as pests and diseases (Wright, 2002; Potvin and Dutilleul, 2009), and these local interactions reduce tree growth across forest ecosystems (Uriarte et al., 2004; Kunstler et al., 2016; Fortunel et al., 2018; Ren et al., 2021, 2023). Neighborhood interactions may exacerbate the impacts of climatic stressors like with water availability and/or freeze cycles, contributing to a more pronounced decline in tree growth (Brooker, 2006; Fichtner et al., 2020; Nemetschek et al., 2024, 2025). For instance, during periods of water stress – be it the dry spells in summer or the freeze-thaw cycles in winter – neighboring trees may intensify competition for soil water and nutrients (Christensen-Dalsgaard and Tyree, 2014; Carnwath and Nelson, 2016). Conversely, neighbors may also foster a microclimate beneath the forest canopy that buffers individual trees from extreme climatic conditions, potentially mitigating the adverse effects on tree growth during periods of macroclimatic stress (de Frenne et al., 2021). Despite these observations, there is a notable gap in research exploring how these observations between climatic stressors and neighborhood dynamics specifically affect tree growth in temperate forest (Rollinson et al., 2016).

Functional traits profoundly influence both the tree intrinsic growth rates (He et al., 2022), and their responses to climate stressors (Uriarte et al., 2016; Bauman et al., 2022) and neighborhood crowding (Uriarte et al., 2010; Fortunel et al., 2016; Kunstler et al., 2016). Species with acquisitive traits – such as low wood density (WD), high specific leaf area (SLA), and high specific root length (SRL) – tend to exhibit rapid growth but are more susceptible to climate stress and competitive pressure from neighbors. Conversely, species characterized by conservative traits – high WD, low SLA, and low SRL – grow more slowly but are generally more resilient to environmental stressors (Bauman et al., 2022; Nemetschek et al., 2024). While SLA and WD are often measured due to their relative ease of assessment, it is their significant ecological implications that make them particularly valuable. These traits shed light on species adaptability and ecological strategies, though they sometimes offer limited explanation for variations in intrinsic growth or sensitivity to climate variations and neighborhood competition (Paine et al., 2015; Kunstler et al., 2016; Poorter et al., 2018; Maréchaux et al., 2020). Recent studies suggested that hydraulic traits may offer a more accurate reflection of tree performance and vulnerability in response to climate and neighbors (Anderegg et al., 2016; Choat et al., 2018; Maréchaux et al., 2020; Oliveira et al., 2021). Traits such as wider and longer xylem vessels facilitate efficient water transport and enhanced photosynthetic capacity (Hacke et al., 2006; Lens et al., 2011), but they also increase the risk of hydraulic dysfunction due to xylem embolism (Davis et al., 1999; Pittermann and Sperry, 2006; Jiménez-Castillo and Lusk, 2013; Gleason et al., 2018). Despite recent evidence on the direct physiological linkages between hydraulic traits and tree growth and

their response to climate stressors and neighbors in tropical forests (Nemetschek et al., 2024, 2025), their implication for temperate tree growth in response to climate stressors and neighborhood competition remain underexplored.

In this study, we analyzed annual tree growth data over 8 years from 593 individuals representing 20 broadleaved deciduous tree species, all of which are representative and common species in temperate old-growth forest in Northeast China. To understand the strategies of these species in response to their environment, we incorporated six functional traits: specific leaf area (SLA), wood density (WD), specific root length (SRL), sapwood hydraulic conductivity, vessel diameter and density. These traits reflect different dimensions of species strategies towards water, carbon, and nutrients. Given the consistent increase in annual temperature, accompanied by growing interannual variability (Figure S1), our aims were to dissect the separate and interactive effects of climate stressors (here intensity of yearly summer water availability and frequency of winter FTC) and neighborhood crowding on tree growth. Additionally, we sought to determine how these functional traits mediate tree responses to these environment effects. Specifically, we aimed to answer the following questions: (1) How do interannual variation in summer water availability and winter FTC influence temperate tree growth? We hypothesize that interannual reductions in summer water availability will suppress tree growth, while interannual increases in winter meteorological freeze-thaw cycles (FTC) frequency will decrease tree growth. We further expect the effect of FTC to exceed that of summer water availability, reflecting stronger hydraulic constraints imposed by winter FTC than by typical growing-season moisture variability in this system. (2) Does neighborhood crowding interact with summer water availability and winter FTC in shaping temperate tree growth? We predict that neighborhood crowding will limit tree growth, and that its interaction with climatic stressors will be negative, jointly exacerbating growth reductions under stressful conditions. (3) How do functional traits influence temperate tree growth and its response to climate and neighborhood crowding? We expect that resource-acquisitive species, characterized by lower wood density, higher SLA/SRL, and larger or more efficient vessels, will grow faster but exhibit greater sensitivity to climatic stressors and neighborhood crowding. Their acquisitive strategy enhances carbon gain and transport efficiency but narrows hydraulic safety margins (i.e., greater cavitation/FTC embolism risk) and reduces structural buffering, leading to steeper growth declines under climatic stress and stronger competitive suppression under high neighborhood crowding.

2. Methods

2.1. Study site

Our research was conducted in the 25-ha permanent forest plot, situated in the Changbai Mountain Nature Reserve (41°43' N ~ 42°26' N, 127°42' E ~ 128°17' E) in Northeast China. This region is characterized by a temperate continental climate, notable for its markedly long and cold winters contrasted with short warm summers. Based on the local meteorological station, the annual mean temperature recorded in the area from 1958–2020 is 3.87 °C, with extremes ranging from a mean maximum of 19.1 °C in August to a mean minimum of –32.8 °C in January. The prolonged sub-zero temperatures, lasting from October to April, account for nearly half of the year, profoundly influences the hydrological and physiological processes of the local vegetation. This extended period of freezing conditions punctuated by thawing events is critical for our study, as it underpins the freeze-thaw cycles that are central to our investigation of their impacts on tree growth. Additionally, the area receives an average annual precipitation of approximately 700 mm, predominantly occurring from June to September (480–500 mm), which contributes to the unique hydrological dynamics of this temperate forest.

In 2004, the 25-ha Changbaishan Forest Dynamics Plot (CFDP) was

established in the core area of the reserve, following the standard field protocol of ForestGEO (<https://www.forestgeo.si.edu>). The CFDP is situated at an average elevation of 801.5 m, with the highest point at 809.5 m and the lowest point at 791.8 m, resulting a maximum elevation difference of 17.7 m. The soil is classified as dark brown forest soil. And soil fertility exhibits an east-to-west gradient, with gradually increasing from the eastern side of the plot toward the western side (Fig. 1a).

This forest is characterized as an old-growth mixed forest dominated by Korean pine (*Pinus koraiensis*) and various deciduous broadleaved tree species, and has remained undisturbed for at least 300 years. The canopy structure is complex, with multi-layered vegetation, providing diverse microhabitats and contributing to structural diversity. The CFDP comprises 52 species, 32 genera, and 18 families (Hao et al., 2008). The dominant species include conifers (*Pinus koraiensis*) and broadleaved species (*Tilia amurensis*, *Quercus mongolica*, *Fraxinus mandshurica*, *Ulmus japonica*, and *Acer mono*). All individuals with a diameter at breast height (DBH) ≥ 1 cm in the CFDP were identified to species, tagged, measured, and mapped. The CFDP has been censused every five years since 2004, with four community surveys conducted in 2004, 2009, 2014 and 2019. The initial census recorded 38,902 individuals, highlighting the high species richness and structural complexity of this old-growth forest.

2.2. Tree growth data

In August 2009, we selected 28 deciduous broadleaved tree species, belonging to 17 genera and 12 families, in the CFDP to monitor tree growth using dendrometer bands. The target deciduous broadleaved

tree species were selected based on the following criteria: (1) They are among the most common species in the study plot, representing 76 % of the total basal area and 56 % of stem density. (2) Their sample sizes are sufficient for reliable statistical analyses, with an average abundance of 970 individuals per species, ensuring adequate representation across species. (3) These species exhibit varying functional traits and ecological strategies, enabling us to assess how species-specific attributes mediate responses to climate stressors (Table 2). By focusing on these species, we ensure a comprehensive yet representative analysis of temperate forest dynamics while maintaining statistical robustness.

To capture the full range of tree developmental stages found in natural forests, individuals of each species with DBH ≥ 5 cm were categorized into nine diameter classes (5–10 cm, 10–20 cm, 20–30 cm, 30–40 cm, 40–50 cm, 50–60 cm, 60–70 cm, 70–80 cm, and over 80 cm). Within each diameter class, we randomly selected 10 individuals per species to ensure sufficient replication and ecological representativeness. In total, dendrometer bands were installed on 927 individual trees across the 28 focal species. After installing dendrometer bands in August 2009 and allowing a six-month acclimation period, we conducted intensive monthly measurements from May to October between 2010 and 2017. These data revealed that the majority of radial growth in temperate trees occurs between early May and late October. Based on these findings, we optimized our sampling schedule starting in 2014, reducing the frequency per season (specially in early May, between August to September, and in late October). This adjustment can maintain data accuracy while improving fieldwork efficiency.

In this study, we excluded (1) trees that died over the census interval; (2) trees whose dendrometer bands were damaged over the census

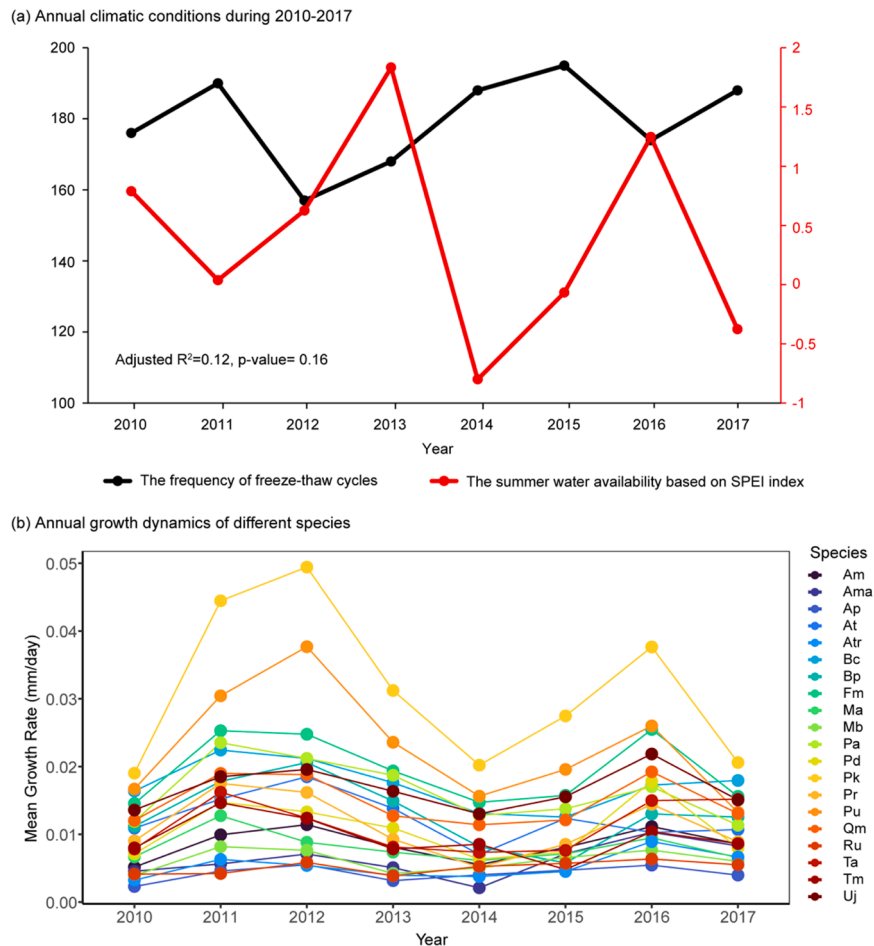


Fig. 1. (a) The frequency of freeze-thaw cycles (FTC) and the summer water availability (WA) based on SPEI index during 2010–2017; (b) Annual growth dynamics of different species. Full species names are listed in Table 1.

interval; and (3) trees from species with fewer than 3 individuals. In total, we analyzed 8 years (2010–2017) of tree growth data for 593 individuals belonging to 20 species (see Table 1, Fig. 1b,c).

We calculated the diameter growth $g_{i,t}$ (mm/day) for each individual i within each census year as:

$$g_{i,t} = \frac{(C_{i,t_o} - C_{i,t_m}) / \pi}{T_{t_o} - T_{t_m}}$$

Where C_{i,t_m} and C_{i,t_o} is the tree circumference in May and October at census year T_t , separately. T_{t_o} and T_{t_m} represent the census date in May and October at census year T_t .

2.3. Freeze-thaw cycles

We defined the frequency of freeze-thaw cycles (FTC) as the number of temperature fluctuations above and below 0°C from October of the previous year to May of the current growing season. Specially, we acquired the hourly air temperature data from October 2009 to October 2017 from the meteorological station, which was located ~4 km from the CFDP (Figure S2). We summed the number of temperature fluctuations above and below 0°C in hourly units from October of the previous year to May of the current growing season (Figure S2). We also calculated the freeze-thaw cycles for different temperature range (i.e. -1 °C ~ 1 °C, -2 °C ~ 2 °C and -3 °C ~ 3 °C), which they all showed strong correlations with the freeze-thaw cycles using the 0°C threshold (Figure S3). We acknowledged that the term freeze and thaw cycles (FTC) is also used in soil science to describe the freezing and thawing of soil water. Although soil freeze-thaw processes can influence trees through their effects on root activity and nutrient dynamics, these processes are strongly modulated by complex factors such as snow cover and soil moisture, and are therefore less direct indicators of stem hydraulic status (Henry, 2007; Kreyling et al., 2012). Accordingly, we used FTC to denote air temperature fluctuations across 0 °C that may directly induce freezing and thawing in stem sap flow, rather than soil freeze thaw processes.

2.4. Water availability

Summer water availability was characterized by the standardized precipitation evapotranspiration index (SPEI) (Vicente-Serrano et al., 2010). The SPEI is a multiscale index calculated from the monthly difference between precipitation and PET (Appendix S1). It uses monthly precipitation data and computes the monthly potential

evapotranspiration (PE) according to the Thornthwaite (1948) equation. Positive and negative SPEI values correspond to relatively wet and dry conditions, respectively. We used the averaged SPEI values of a time window covering 5 months, which can be used to characterize the water availability during the growing season (i.e. from May to October). Although this period extends beyond the meteorological summer (June–August), it captures the ecologically relevant window of tree growth in Northeast China. Therefore, to ensure consistency with the winter freeze-thaw cycle analysis and to provide a complete seasonal framework for evaluating climatic effects on tree growth, we defined the period from May to October as the growing-season. Previous studies have indicated that drought stress, particularly during the early growing season (May–July), significantly constrains tree growth in temperate forests (Wang et al., 2020; Yuan et al., 2021; Gao et al., 2022; Zhu et al., 2023). To evaluate this possibility, we assessed alternative SPEI intervals, including the May to July period. Our analyses revealed no statistically significant relationship between SPEI during this early growing season and tree growth (Figure S4). Therefore, we retained SPEI from May to October to characterize summer water availability in this study.

2.5. Neighborhood crowding

To evaluate the effect of neighbors on tree growth, we calculated a neighborhood competition index (NCI) (Canham et al., 2004) for each focal tree i based on the size and distance of its $N(i)$ neighbors within a 20 m radius (Wang et al., 2010):

$$NCI_i = \sum_{j \in N(i)} \frac{DBH_j^2}{dist_{ij}^2}$$

Where DBH_j is the size of the neighbor tree j and $dist_{ij}$ is the distance between focal tree i and its neighbor j .

The NCI framework integrates both the size and spatial proximity of neighboring trees relative to each focal tree, thereby capturing the density-dependent competition across the plot. Since the whole CFDP was censused at two dates overlapping the dendrometer surveys (2009 and 2014), we calculated NCI at two periods (between 2009 and 2013, and between 2014 and 2017).

2.6. Functional traits

For the 20 focal species, we used standard protocols (Tyree et al.,

Table 1

List of the 20 focal tree species with the number of individuals sampled, DBH range and mean growth rate with standard deviation.

Species	Abbreviation	Family	Genus	Ind. Num.	DBH range (cm)	Growth [mm/day]
<i>Populus koreana</i>	Pk	Salicaceae	Populus	4	69.7[113.7,45.3]	0.0313±0.0124
<i>Populus ussuriensis</i>	Pu	Salicaceae	Populus	11	94.5[142,40.4]	0.0229±0.0128
<i>Fraxinus mandshurica</i>	Fm	Oleaceae	Fraxinus	55	40.2[94.0,6.7]	0.0194±0.0116
<i>Betula costata</i>	Bc	Betulaceae	Betula	8	31.7[61.4,10.7]	0.0173±0.0118
<i>Ulmus japonica</i>	Uj	Ulmaceae	Ulmus	54	36.0[86.3,5.4]	0.0167±0.0122
<i>Phellodendron amurense</i>	Pa	Rutaceae	Phellodendron	21	22.4[35.8,13.0]	0.0163±0.0094
<i>Quercus mongolica</i>	Qm	Fagaceae	Quercus	64	42.4[106.0,7.1]	0.0148±0.0103
<i>Betula platyphylla</i>	Bp	Betulaceae	Betula	30	26.3[49.9,9.4]	0.0130±0.0091
<i>Acer tegmentosum</i>	At	Aceraceae	Acer	15	9.3[14.1,5.1]	0.0123±0.0079
<i>Tilia amurensis</i>	Ta	Tiliaceae	Tilia	50	43.3[111,5.6]	0.0112±0.0102
<i>Padus racemosa</i>	Pr	Rosaceae	Padus	21	14.0[26.5,5.0]	0.0111±0.0131
<i>Populus davidiana</i>	Pd	Salicaceae	Populus	9	19.4[30.1,5.7]	0.0107±0.0097
<i>Tilia mandshurica</i>	Tm	Tiliaceae	Tilia	45	26.1[51.8,5.5]	0.0094±0.0072
<i>Acer mono</i>	Am	Aceraceae	Acer	42	23.7[45.6,5.2]	0.0085±0.0073
<i>Maackia amurensis</i>	Ma	Leguminosae	Maackia	18	15.5[27.8,6.3]	0.0081±0.0061
<i>Acer mandshuricum</i>	Ama	Aceraceae	Acer	28	16.8[34.9,5.1]	0.0063±0.0057
<i>Malus baccata</i>	Mb	Rosaceae	Malus	48	18.4[42.5,5.3]	0.0061±0.0054
<i>Acer triflorum</i>	Atr	Aceraceae	Acer	31	16.2[33.0,5.4]	0.0053±0.0056
<i>Rhamnus ussuriensis</i>	Ru	Rhamnaceae	Rhamnus	15	10.7[18.5,6.4]	0.0051±0.0039
<i>Acer pseudo-sieboldianum</i>	Ap	Aceraceae	Acer	24	13.0[25.5,5.1]	0.0042±0.0038

1993; Cornelissen et al., 2003) to measure six functional traits at the stem, root and leaf levels that represent important dimensions of species ecological strategies towards water, carbon and nutrients (Table 2, Figure S5). In this study, we selected mature individuals to ensure trait comparability across species, which is a widely accepted approach in trait-based ecological studies because interspecific trait differences among mature individuals are generally more stable and representative of species-level strategies (Ren et al., 2021; Umaña et al., 2025). Specifically, for each species, we selected six individuals that were as close to maturity as possible while also representing a range of sizes to capture natural variability within mature cohorts (Table S6).

Mean vessel diameter (D_v) and vessel density (VD) are anatomical traits that relate to water transport from roots to leaves (Hacke and Sperry, 2001). To minimize variation in sampling height and its potential confounding effects on hydraulic traits, all branches were sampled within 5 m above the ground. The stem segments used for hydraulic conductivity measurements were used for wood anatomical analysis. For each segment, we cut a 20- μm -thick cross section using the sliding microtome (Model 2010–2017, Shanghai Medical Instrument Corp., Shanghai, China) and stained it in a 0.1 % toluidine blue solution (~ 30 min) to increase visual contrast. The preparation was observed under a magnification of $40\times$ using a digital camera mounted on a light microscope (Leica ICC50, Wetzlar, Germany). We then used Image-J software to measure vessel lumen areas. D_v was calculated for each segment ($n > 100$ conduits) and a grand mean was determined for each species ($n = 6$ segments). We calculated VD as the number of vessels per studied xylem area (Tyree and Ewers, 1991).

Sapwood-specific hydraulic conductivity (K_s) reflects xylem water transport efficiency (Hao et al., 2013). For each tree, we collected six straight sun-exposed branches (~ 100 cm) with a diameter of about 8 mm in the early morning. The branches are immediately stored in large plastic bags with damp paper towels to prevent desiccation. In the laboratory, we cut out a 20 cm stem segment (~ 20 cm) from each branch under water. Then we measured the rate of liquid flowing (0.20 μm filtered and degassed 20 mmol L^{-1} KCl solution) through the branch under unit pressure and unit length as the xylem hydraulic conductivity (K_h). After the measurement, we cut the branches to make 20- μm -thick slices, then dyed the slices and placed it on a scanner to scan and save the pictures. We used Image-J software (US National Institute of Health, Bethesda, MD, USA) to measure the sapwood area. We calculated K_s as

K_h divided by the cross-sectional sapwood area (Hao et al., 2013).

Wood density (WD) reflects structural support and defense (Chave et al. 2009). Due to strict conservation regulations within the Changbai Mountain Nature Reserve, the use of increment borers to extract wood cores was prohibited. To obtain wood density data, we selected an alternative managed forest site approximately 10 km away from the CFDP, where sampling activities are permitted. This site was selected for its ecological similarity to the study plot, both featuring a deciduous broadleaved Korean pine (*Pinus koraiensis*) mixed forest with comparable climatic condition, soil type, pH, and nutrient profiles. These similarities help minimize potential biases when applying wood density data from the alternative site to the study plot. At this alternative site, we collected 10 stem cross-sections for each species. We measured the fresh volume using the water displacement method, and weighed the dry mass after drying in the oven at 100°C for 48 h. We calculated WD as the oven-dried mass divided by fresh volume (Cornelissen et al., 2003).

Specific root length (SRL) reflects soil water and nutrients acquisition (Kramer-Walter et al., 2016). We selected six mature individuals per species in the CFDP, dug out the sub-roots that stemmed along the main root, and used a shovel to collect a root sample with the surrounding soil. We cleaned the soil particles around the roots with running water. We picked fine roots (diameter < 2 mm) under a dissecting microscope and scanned it using a portable scanner (Canon LiDE 110, Tokyo, Japan). The scanned picture was used to measure root length with WinRHIZO software (Régent Instruments, QC, Canada). We oven-dried root samples at 80°C for at least 48 h before weighing. We calculated SRL as root length divided by root dry mass (Cornelissen et al., 2003).

Specific leaf area (SLA) reflects light acquisition and photosynthesis (Wright et al., 2004). We collected 10 fully-expanded, intact canopy and fully exposed to sunlight leaves from the 6–12 largest individuals of each species in the CFDP. We scanned each leaf with a portable scanner before oven-drying at 60°C for 48 h to get leaf dry mass. The leaf area was measured with WinFLOIA software (Régent Instruments, QC, Canada). Here, we calculated SLA as leaf area divided by leaf dry mass (Cornelissen et al., 2003).

2.7. Statistical analyses

Firstly, to avoid the collinearity among predictor variable, we evaluated the potential collinearity among predictor variables by calculating the variance inflation factor (VIF). The VIF values were all below 2 (ranging from 1.04 to 1.95) (Table S2), indicating negligible collinearity among predictor variables. Then, we constructed a hierarchical Bayesian model to investigate the impacts of freeze-thaw cycles, summer water availability, and neighborhood interactions on tree growth. Since we found no significant relationships between the two climate stressors (Fig. 1a), we combined them in the model. The first-level regression assesses the overall tree growth response to climate stressors, neighbors and their interactions through the hyper-distribution, whereas the second-level regression examines how species traits mediate tree growth responses to the model covariates.

The first-level (community-level) regression modeled the expected log-transformed annual diameter growth $g_{t,s,i}$ of an individual stem i of species s between censuses t and $t + 1$ as:

$$E[\log(g_{t,s,i})] = \beta_{1s} + \beta_{2s} \log(DBH_{t,s,i}) + \beta_{3s} WA_t + \beta_{4s} FTC_t + \beta_{5s} \log(NCI_{t,s,i}) + \beta_{6s} WA_t * \log(NCI_{t,s,i}) + \beta_{7s} FTC_t * \log(NCI_{t,s,i}) + \delta_i$$

where, for a given stem i of species s , $DBH_{t,s,i}$ is the stem diameter at 1.3 m height at census t , WA_t is the summer water availability during the growing season prior census t , FTC_t is freeze-thaw cycle from October to May prior to growing season prior census t , and $NCI_{t,s,i}$ is the neighborhood crowding index of the focal stem i of species s for the period encompassing census t . δ_i is the random effect of the 20 m x 20 m

Table 2
List of studied functional traits.

Trait (Abbreviation)	Unit	Description	Function	Mean [min, max]
Mean vessel diameter (Dv)	μm	Mean vessel diameter	Water hydraulic transport	31.6 [18.3, 60.8]
Vessel density (VD)	no. mm^{-2}	Number of vessels per area.	Water hydraulic transport	240.3 [43.5, 504.1]
Native sapwood-specific hydraulic conductivity (Ks)	$\text{kg m}^{-1} \text{s}^{-1} \text{Mpa}^{-1}$	K_h/XA , where “ K_h ” is the stem hydraulic conductivity, and “ XA ” is the xylem area (mm^2).	Water hydraulic transport	3.3 [1.3, 6.9]
Wood density (WD)	g cm^{-3}	Wood dry mass dived by volume	Structural support, hydraulic capacity, defense	0.5 [0.3, 0.7]
Specific root length (SRL)	g cm^{-1}	Fine root dry mass dived by length	Nutrient absorption	3483.8 [1785.8, 5367.3]
Specific leaf area (SLA)	g cm^{-2}	Leaf dry mass dived by area	Light interception	219.7 [147.2, 312.8]

quadrat where each individual is located.

The second-level regression evaluates how species traits mediate tree growth responses to summer water availability, FTC, neighborhood crowding and their interactions. For a tree of species s , the species-specific regression coefficient β_{cs} in the first level is modelled as a function of species trait as follows:

$$\beta_{cs} = \beta_{c,0} + \beta_{c,1}$$

where $\beta_{c,0}$ is the community-level intercept term, $\beta_{c,1}$ are the departures from the grand slope of the corresponding model covariates for an increase of one standard deviation in the value of trait F_s of species s .

To ensure direct comparison of parameter estimates and model convergence, all covariates were standardized to enable direct comparisons of the relative importance of different predictors. *DBH* and *NCI* were log-transformed prior to standardization. Considering that species may vary in tree size range and experience different ranges of neighborhood crowding, *DBH* and *NCI* were standardized by subtracting species-specific means and dividing by species-specific standard deviations. Functional traits were log-transformed to prevent the tail of their skewed distribution from dominating the model fit. Subsequently, they were scaled by subtracting mean and dividing by one standard deviation across all species. Other continuous variables were standardized by subtracting mean and dividing by one standard deviation across the dataset.

We fit separate models for each of the six traits. We used diffuse prior distributions for all parameters and estimated the parameters using the R2jags package (Su and Yajima, 2015). We ran three parallel chains with different initial values, and used Gelman and Rubin's convergence diagnostics (with a threshold value <1.1) to detect parameter convergence (Plummer et al., 2006). Hereafter we present parameters estimates as the median, 2.5 % and 97.5 % quantiles (m [q_{2.5}, q_{97.5}]). We evaluated the goodness of fit for each model using standard major axis regression to retrieve the R^2 between observed growth and modeled growth simulated from the posterior samples (Table S2).

To comprehensively address potential confounding factors in our data, we implemented a series of complementary analytical approaches. First, to address potential size-related effects due to the wide *DBH* variation in our dataset, we performed additional stratified analyses by dividing individuals into small and large size classes based on species-specific median *DBH* (50th percentile). Then, to fully explore the potential effects of individual climate variables, we further replaced *SPEI* with growing-season temperature and precipitation as independent predictors. Given that *FTC* frequency is a complex compound indicator influenced by various climatic factors, we separately quantified freezing events (runs $< 0^\circ\text{C}$) and thawing events (runs $> 0^\circ\text{C}$) for October–April. To identify the primary contributors of *FTC* frequency, we fitted two simple linear regression models relating *FTC* frequency to each component (e.g., freezing events and thawing events). Finally, we calculated simpler and more direct winter-temperature variables (e.g., mean winter daytime temperature and warm-day frequency ($T_{\max} > 0^\circ\text{C}$)) and re-fitted a Bayesian hierarchical growth model that included *DBH*, competition (*NCI*), *FTC* ($\pm 0^\circ\text{C}$ crossings), mean winter daytime temperature, warm-day frequency ($T_{\max} > 0^\circ\text{C}$), and *SPEI*, plus *NCI* interactions and random intercepts for species and plot. We then compared the full model with nested variants that removed one predictor at a time and used Deviance Information Criterion (ΔDIC) as a primary criterion for model comparison (lower values indicate better fit; larger positive ΔDIC reflects greater information loss). As these additional analyses yielded results highly consistent with the main model, for clarity and conciseness, we present only the full model results in the main text. All supplementary analyses and detailed results are provided in the Appendix.

All analyses were conducted in R 4.4.1 (R Development Core Team; <http://r-project.org>).

3. Results

According to our comparative analysis of annual tree growth data, 2014 exhibited significantly lower growth compared to 2011, 2012, 2016, and 2017, but growth rates were similar to those observed in 2010 and 2015 (Figure S6).

Overall, we found a negative effect of *FTC* on tree growth in our study site, whereas summer water availability did not significantly affect growth (Fig. 2, Table S3). At the community level, tree growth declined with increasing frequency of *FTC* ($\beta_{4,0}$: $-0.08[-0.12, -0.05]$). At the species level, half of the species assessed exhibited reduced growth rates corresponding to more frequent of *FTC*, while the other half of the species did not display a marked response strong to the *FTC* (β_{4s} parameter in Table S4). Regarding summer water availability, neither community level nor species specific analyses showed a substantial influence on tree growth ($\beta_{3,0}$ and β_{3s} parameter in Table S3 and Table S4). This pattern was consistent across both small and large trees, with individuals in different size classes similarly exhibiting reduced growth under increased freeze-thaw frequency but showing no significant response to growing-season water availability (Figure S7). The additional analysis further demonstrated that *FTC* contributes non-redundant information beyond simpler warming metrics. Whereas increases in *FTC* frequency were associated with both more frequent freezing and thawing events (Figure S8), its removal from the full model resulted in the largest DIC penalty ($\Delta\text{DIC} = +207.6$), far exceeding those of mean winter temperature ($\Delta\text{DIC} = +17.4$), warm-day frequency ($\Delta\text{DIC} = +86.9$), and *SPEI* ($\Delta\text{DIC} = +3.6$) (Table S7).

Consistent with our predictions, we found that tree growth decreased with neighborhood crowding at the community level ($\beta_{5,0}$: $-0.1[-0.17, -0.03]$). Size-specific analyses further showed that neighborhood crowding had a significant negative effect on small trees, while the effect was not significant for large trees (Figure S7). At the species level, neighbors clearly limited tree growth in 13 species, while increasing tree growth for 2 species, such as *Fraxinus mandshurica* and *Populus koreana* (β_{5s} parameter in Table S4). However, we did not find evidence that neighbors interact with water availability or freeze-thaw cycles in shaping tree growth at either the community or species level ($\beta_{6,0}$, $\beta_{7,0}$, β_{6s} , β_{7s} parameter in Table S3 and Table S4).

Finally, we demonstrated that fast-growing species were associated with higher xylem hydraulic efficiency ($\beta_{1,1}$: $0.49[0.30, 0.68]$), lower wood density ($\beta_{1,1}$: $-0.36[-0.61, -0.11]$) and lower specific leaf area ($\beta_{1,1}$: $-0.43[-0.65, -0.21]$) (Fig. 3, Table S3). Functional traits also mediated the tree growth response to freeze-thaw cycles and neighbors (Fig. 4). In particular, species with lower xylem hydraulic efficiency ($\beta_{3,1}$: $-0.05[-0.08, -0.01]$), greater wood density ($\beta_{3,1}$: $0.05[0.01, 0.09]$) and greater specific leaf area ($\beta_{3,1}$: $0.04[0.003, 0.07]$) were less sensitive to the negative effect of freeze-thaw cycles. Conversely, species with higher xylem hydraulic efficiency ($\beta_{4,1}$: $0.08[0.01, 0.16]$) were less sensitive to the negative effect of neighbors (Fig. 4, Table S3).

4. Discussion

Identifying the separate and interactive effects of major factors driving tree growth is crucial for accurately predicting and managing forest dynamics and functions with global change. In our study, we dissected the separate and interactive effects of summer water availability, freeze-thaw cycles, and neighborhood crowding on tree growth. Additionally, we explored how these influences vary according to species functional traits that govern strategies for water, carbon, and nutrients utilization. This comprehensive analysis provides insights into the mechanisms through which environmental variables and intrinsic biological traits shape tree growth, offering a nuanced view of forest resilience and adaptability to climate variability.

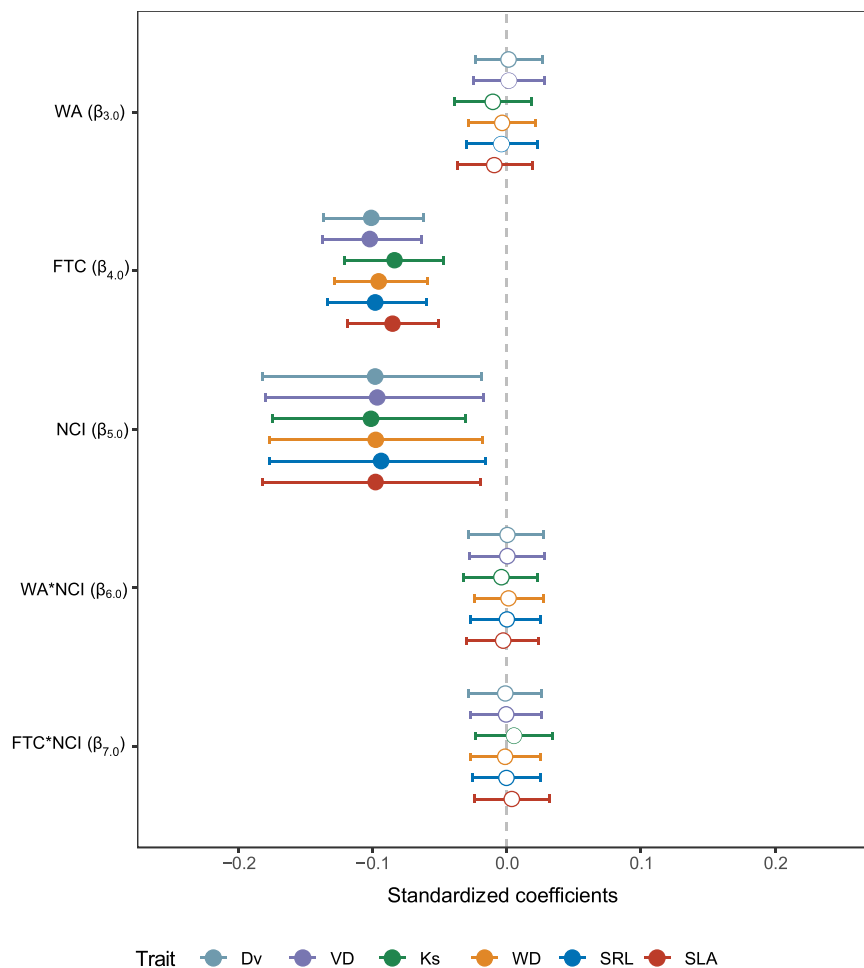


Fig. 2. The effects of the summer water availability (WA), winter freeze-thaw cycles (FTC), neighborhood crowding (NCI) and their interaction on the tree growth rate. Coefficients are shown for each of the six trait models. Each point is the posterior medians and each line segment is the 95 % credible intervals respectively. Filled circles represent significant effects (i.e. when 95 % credible intervals are not encompassing zero).

4.1. Climate and neighbors act largely independently on tree growth

Across 20 common temperate tree species in the CFDP, the frequency of freeze-thaw cycles significantly hinders tree growth, while the summer water availability does not notably affect tree growth. Our findings contrast with the global reports of reduced tree growth due to water scarcity (Martin-Benito et al., 2018; Zuidema et al., 2022). This discrepancy may stem from our study site's location in a humid cold forest, characterized by a typical temperate continental climate with monsoonal influence, featuring long, cold winters and short, rainy summers. The region receives abundant precipitation (averaging 700 mm annually), with over 70 % falling in July and August (Yang, 1981), ensuring sufficient water availability during the growing season. This inference is further supported by our additional analyses using growing-season temperature and precipitation as independent predictors, which showed no effect of precipitation on tree growth. Although summer temperature exhibited a negative effect on tree growth, its positive correlation with precipitation suggests that increased rainfall may alleviate the heightened evaporative demand associated with higher temperatures (Figure S9; Figure S10). This compensatory interaction likely reduced overall water stress, thereby diminishing the drought signal captured by SPEI and explaining its non-significant association with tree growth. Additionally, the severity of droughts during our study period was close to the regional average (Ma et al., 2023), indicating that the studied species were generally not strongly limited by water during typical summer droughts. These results

indicate that, under current climatic conditions, summer drought is unlikely to be a major growth-limiting factor in this region. However, it is plausible that more severe or prolonged water-stress conditions would hinder tree growth (Anderegg et al., 2020), especially considering the predicted increase in aridification in temperate forests due to ongoing climate change (Overpeck and Udall, 2020).

Consistent with our predictions, the severity and frequency of FTC were closely linked to reduced tree growth, suggesting that climate variations outside the growing season can also play a critical role in determining temperate tree growth. In humid temperate forests, FTC can significantly impact tree growth by compromising hydraulic function, often posing a greater threat than drought (Niu et al., 2017). The repeated freezing and thawing of xylem sap during winter can induce embolism formation, particularly in broadleaved species with larger vessel diameters, where FTC can cause up to a 50 % loss in xylem hydraulic conductivity. If trees fail to restore their hydraulic function before the growing season, their ability to transport water efficiently is severely impaired, ultimately limiting growth (Pittermann and Sperry, 2006; Choat et al., 2011). Furthermore, to cope with FTC-induced embolism, tree species employ different strategies. Some invest in the production of earlywood with large conduits to enhance water transport capacity, while others rely on root pressure in early spring to refill embolized vessels. Both strategies require substantial allocation of non-structural carbohydrates (NSCs), which are also essential for biomass accumulation. (Yin et al., 2022). While previous research on temperate forests has acknowledged the impact of non-growing season

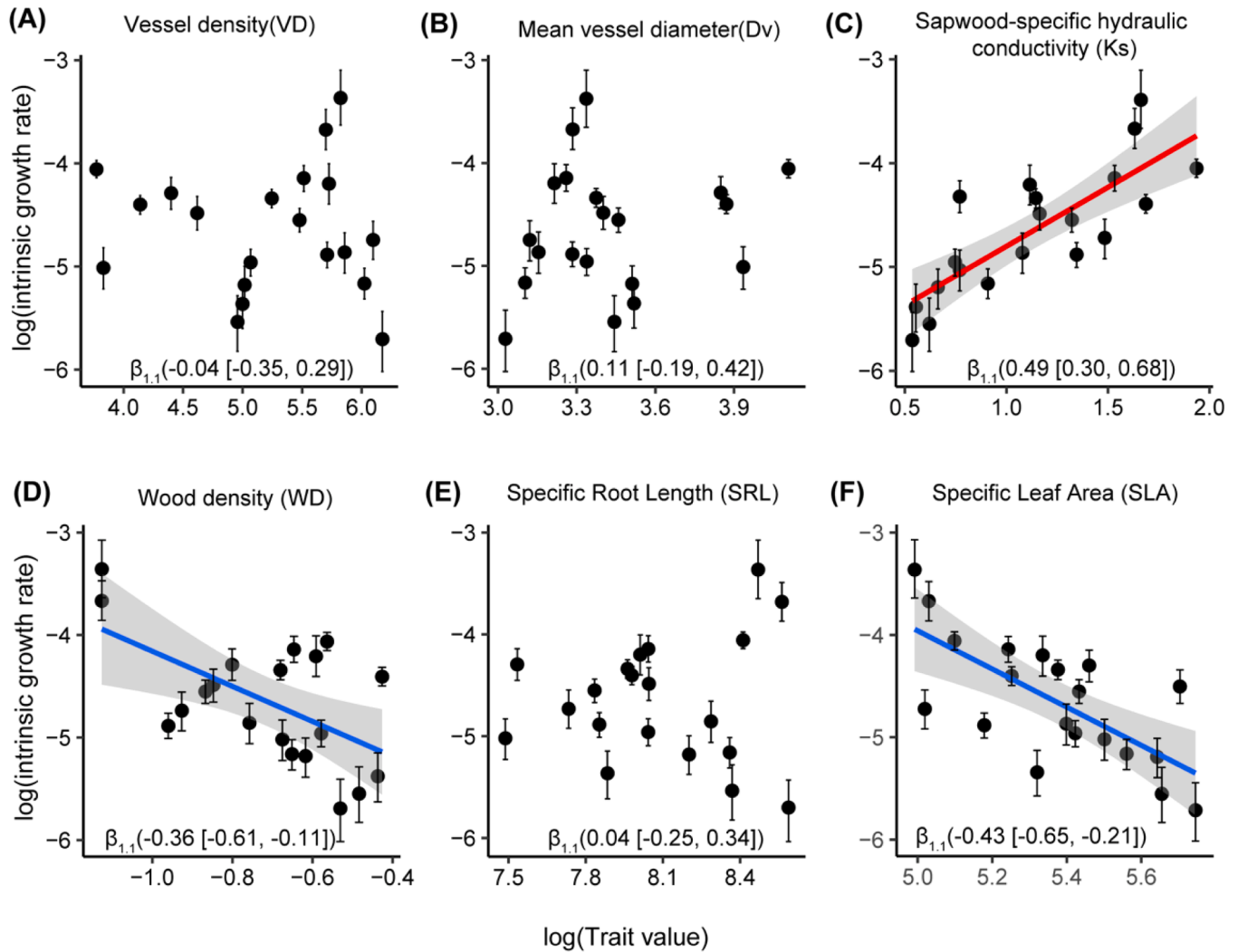


Fig. 3. The traits-growth relationship across all 20 species in our study. Each point is the posterior means and each line segments is the 95 % credible intervals respectively. Circles are species, placed at the median of their corresponding coefficient posteriors. Vertical and horizontal bars are 95 %-highest posterior density interval for the corresponding coefficients. Red and blue regression lines indicate positive and negative correlations. Community-level median, lower and upper 95 % are in the bottom of the figures.

climate, the focus has predominantly been on the effects of anomalously cold or warm years (D'Orangeville et al., 2016; Harvey et al., 2020). Leveraging high-resolution temperature records, our study shifts the focus to intra-seasonal variability, specifically freeze-thaw cycles (FTC) that are not captured by seasonal means. We further show that the increased FTC frequency is driven by a combination of more frequent freezing and thawing events. This pattern aligns with additional physiological pressure that FTC may exert on tree growth. Specifically, while cold extremes are directly tied to frost injury, warmer winter days can elevate respiratory carbon losses and reduce chilling accumulation, potentially delaying or impairing spring cambial reactivation (Delpierre et al., 2019; Wang et al., 2025). With intensifying climate anomalies, the frequency of FTC is likewise expected to increase in temperate regions (Wu et al., 2025). Explicit consideration of FTC dynamics is therefore essential not only for climate change impact assessments but also for maintaining forest ecosystem functioning. Moreover, incorporating physiological indicators such as stem sap flow and cell water potential would offer a more direct and mechanistic understanding of tree responses to freeze-thaw stress.

Moreover, our results illustrate that tree growth declines with increasing neighborhood crowding in the CFDP and smaller trees exhibited a stronger sensitivity to neighborhood crowding than large

tree. This finding is consistent with previous studies conducted in tropical, subtropical and temperate forests (Lasky et al., 2014, 2015; Wills et al., 2016; Zhang et al., 2017; Fortunel et al., 2018; Rozendaal et al., 2020; Ren et al., 2021, 2023). The negative effect of neighborhood crowding suggests a prevalent role for biotic interactions, either directly via competition for resources or indirectly via shared natural enemies, in shaping tree growth and species dynamics (Fortunel et al., 2016; Adler et al., 2018). Interestingly, some species like *Fraxinus mandshurica* and *Populus koreana* had higher growth under increased neighborhood crowding, as was shown in previous studies in different ecosystems (Lasky et al., 2015; Fortunel et al., 2018). This pattern could be due to species-specific adaptations (Bertness and Callaway, 1994) or beneficial microclimatic conditions created by neighbor presence (Coates et al., 2013). Additionally, contrasting effects of conspecific vs heterospecific neighbors or trait differences between neighbors may contribute to this phenomenon, as increased ecological dissimilarity among neighbors may benefit tree growth (Fortunel et al., 2016; Kunstler et al., 2016; Nemetschek et al., 2025). Future study that partition the relative contributions of conspecific vs heterospecific neighbors and consider the trait-based neighborhood interaction could provide further insight into the ecological processes shaping tree growth (Kalyuzhny et al., 2023).

Contrary to our initial hypothesis, we found that the effect of

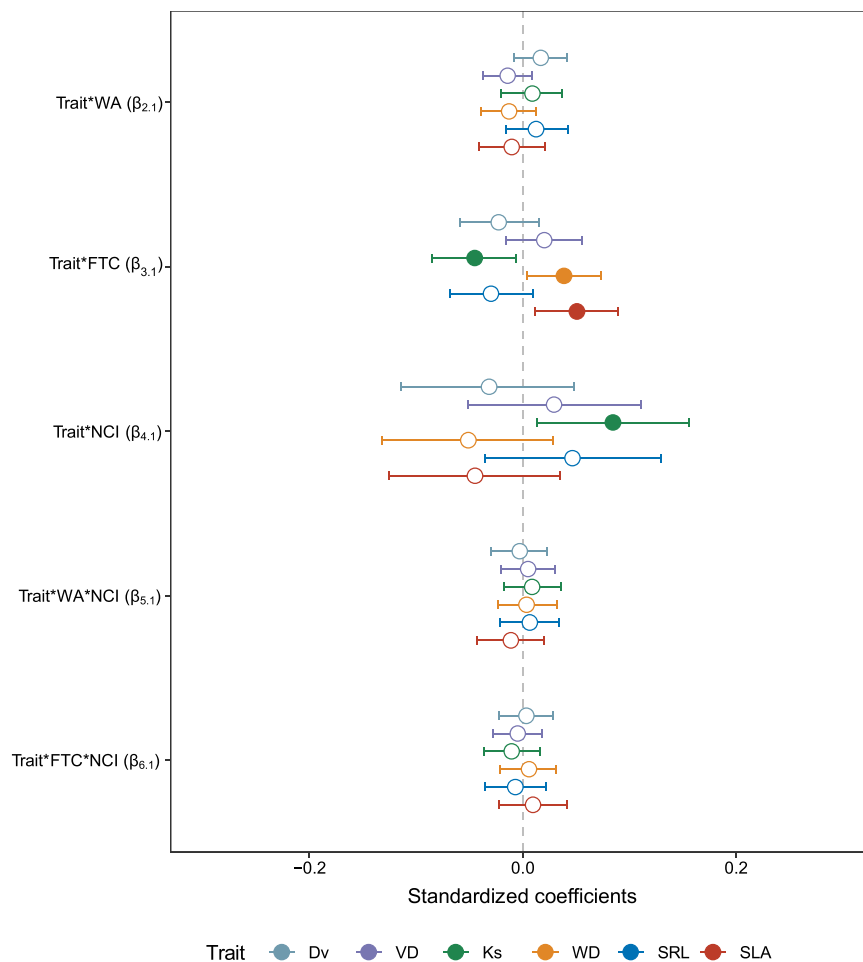


Fig. 4. The trait-mediated tree growth sensitivity to summer water availability (WA), winter freeze-thaw cycles (FTC), neighborhood crowding and their interaction. Each point is the posterior medians and each line segments is the 95 % credible intervals respectively. Filled circles represent significant effects (i.e. when 95 % credible intervals are not encompassing zero).

neighborhood crowding and climate stressors on tree growth largely operate independently rather than interactively in the CFDP. Previous studies also suggested that abiotic filtering and biotic interactions largely act independently in shaping the tree performance (but see (Sanczuk et al., 2023; Nemetschek et al., 2024)). Consistent with tropical-forest evidence, Nemetschek et al. (2024) found that climate anomalies and neighborhood crowding interact to shape tree growth, with interactions that can attenuate the separate negative effects of heat and drought stress and crowding. Our results further supported this perspective by considering the combined effect of climate-related abiotic filtering and biotic interactions on tree performance (Fisher et al., 2018; Fortunel et al., 2018; Nemetschek et al., 2024). The independent roles of abiotic filtering and biotic interactions highlights the importance of considering multiple ecological processes in evaluating and predicting temperate forest dynamic under climate change. In this study, we focused on stems with DBH ≥ 5 cm, and these established stems may be able to tolerate the combined effects of climate stress and biotic interactions. However, younger and less established seedling may exhibit different vulnerabilities (Xu et al., 2022), suggesting a need for future research to explore growth responses across different developmental stages within temperate forests.

4.2. Species traits mediate tree growth response to climate and neighbors

Our study reinforces the significant role functional traits play in species intrinsic growth rate in our study site, providing further evidence

that they can be effective proxies for assessing species ecological strategy and performance (Poorter et al., 2008; Rüger et al., 2012). As expected, species with light wood density (WD) and higher hydraulic efficiency (Ks) exhibited faster growth, benefiting from lower construction costs and enhanced ability to acquire water and carbon (Santiago et al., 2004; Zhang and Cao, 2009; Fan et al., 2012). Unexpectedly, species with higher specific leaf area (SLA), typically linked to lower investment in leaf dry matter and higher photosynthetic efficiency (Wilson et al., 1999; Wright et al., 2004), showed lower average growth in our study. This negative association between SLA and growth rate had been observed in several studies (Poorter et al., 2008; Uriarte et al., 2016; Muscarella et al., 2018; Ren et al., 2021), although the mechanisms underlying this negative relationship remain unclear. One possible explanation may be that SLA is an integrative trait, reflecting various leaf characteristics (e.g., leaf tissue density, leaf area, and leaf thickness) that may not uniformly respond to environmental pressures, potentially explaining the observed inverse relationship between SLA and tree growth.

Furthermore, our findings indicate that species with more conservative strategies, such as higher wood density and lower hydraulic efficiency, were less impacted by the frequent of FTC. This pattern is consistent with the xylem safety–efficiency trade-off and the broader conservative–acquisitive spectrum, in which traits that enhance hydraulic safety buffer FTC stress at the expense of growth. Mechanistically, safer xylem anatomy (smaller conduits, denser wood, safer pit membranes) limits cavitation formation and spread, while tighter winter

water-potential control and lower metabolic demand help preserve NSC for repair (Hacke and Sperry, 2001; Niu et al., 2017; Groover et al., 2025). Together, these features allow conservative species to conserve resources more effectively and recover more quickly from freeze-thaw-induced hydraulic disruption. Given ongoing and projected increases in temperature and precipitation variations, we anticipate that more frequent droughts and freeze-thaw cycles will have compounded effects on tree growth, potentially leading to substantial changes in forest composition, favoring species with conservative strategies towards water, carbon and nutrients. Moreover, while previous studies found that relatively easy-to-measure traits can mediate tree growth and survival response to neighborhood crowding in forest ecosystems worldwide (Kunstler et al., 2016; Fortunel et al., 2018; Muscarella et al., 2018), our study found no effect of these traits on tree growth response to neighbors. However, we demonstrated that species with higher xylem hydraulic efficiency grew faster in crowded neighborhoods, suggesting that hydraulic traits can capture tree growth response to neighbors when water limitation dominates neighbor interactions (Fan et al., 2012; Eller et al., 2018; Nemetschek et al., 2024, 2025). Lower xylem hydraulic efficiency provides higher resistance to embolism, but may also limit metabolic rate by increasing resistance to water transport during the growing season, thereby resulting in a disadvantage in crowded neighborhoods (Hacke et al., 2006; Pratt et al., 2007). Overall, these findings support the notion that the safety-efficiency trade-off in xylem hydraulics underpins species-specific growth responses to both climatic and biotic factors in temperate forests. This trade-off facilitates the coexistence of diverse functional strategies, enabling species to allocate resources differently between resistance to abiotic stressors and growth under biotic interactions. Such divergent strategies may further contribute to the maintenance of species diversity in temperate forests. Nevertheless, the underlying mechanisms driving species-specific sensitivities remain incompletely understood, particularly with respect to adaptive hydraulic responses (Li et al., 2024; Liu et al., 2024), carbohydrate storage strategies (Medeiros et al., 2025), and root plasticity (Harrison Day et al., 2024). Additionally, the lack of growth duration data limits our ability to fully assess FTC effects on interspecific growth variation (Zhang et al., 2025). Future research should integrate physiological trait measurements, dendrometer-based growth tracking, and phenological monitoring to better capture species-specific adaptations and seasonal growth dynamics.

Climate change will pose greater pressure on the potential of forest in carbon sequestration by influencing forest dynamics (McDowell et al., 2020). Our findings suggest that greater attention should be paid to non-growing season climatic stressors and local neighborhood interactions to understand how multiple drivers will impact temperate forest dynamics and subsequent carbon sink. While our study focused on dominant species, future research assessing all species present in the plot would provide a more comprehensive understanding of how temperate forests respond to climate change. Additionally, while we mainly focused on the species-level functional traits of above-ground organs, exploring more belowground traits and individual-level traits may offer complementary insight into the intraspecific trait variation and how traits mediate plant responses to climate change. Another important direction for future research is the integration of high-resolution microclimate data with complementary environmental variables, such as soil moisture and atmospheric vapor pressure deficit (VPD) (Zellweger et al., 2020). Such an integrated framework is essential for capturing the fine scale environmental variability that governs tree performance and species interactions under a changing climate.

5. Conclusion

In this study, we demonstrated that climate and neighbors acted mainly independently in driving tree growth in a temperate forest. In particular, freeze-thaw cycles and neighborhood crowding consistently reduced tree growth, while summer water availability had no effect.

Furthermore, we found that some functional traits determined species intrinsic growth, and hydraulic traits can mediate both the influence of freeze-thaw cycles and neighbors on tree growth. Overall, our study improved our understanding of the drivers underlying temperate tree growth, thereby providing important insight into better forecasting temperate forest dynamics under different climate trajectories.

CRediT authorship contribution statement

Jing Ren: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Shuai Fang:** Writing – review & editing, Writing – original draft, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Guangyou Hao:** Writing – review & editing. **Fei Lin:** Investigation, Funding acquisition. **Ji Ye:** Investigation. **Zhanqing Hao:** Investigation. **Xugao Wang:** Writing – review & editing, Supervision, Investigation, Funding acquisition. **Claire Fortunel:** Writing – review & editing, Writing – original draft, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.agrformet.2025.110954.

Data availability

Data will be made available on request.

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